

Week 1 Cyber Threat Intelligence Fundamentals

Project topic: Remote PowerShell Attacks via WinRM

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1. Introduction to CTI Concepts

Our project examines how PowerShell Remoting can be used during an attack. This Windows feature allows an authorized user to run commands on another computer. In the scenario we study, the connection is made through Windows Remote Management (WinRM).

PowerShell Remoting is commonly used by system administrators, so a remote session is not automatically suspicious. The problem arises when someone uses an account without authorization or misuses access they already have. To understand such activity, an investigator needs to know which account connected, where the connection came from, which computer was accessed and what commands were run.

In the first week, we focus on the terms needed to describe this scenario and on the types and sources of threats related to it.

2. Glossary of Key CTI Terms

PowerShell Remoting - a feature that allows PowerShell commands to run on a remote computer. It is the main functionality examined in our project.

WinRM (Windows Remote Management) - a Windows service that supports remote management. In our project, it provides the connection to the remote computer.

PSSession - a PowerShell session that can maintain a connection to a remote computer. Several commands can be run within the same session.

Invoke-Command - a PowerShell cmdlet used to run commands on another computer. It can use a temporary connection or an existing PSSession.

Enter-PSSession - a PowerShell cmdlet that starts an interactive session with a remote computer. The user can then enter commands within that session.

Script Block Logging - a feature that can record PowerShell script content when enabled. These records can help an investigator see what PowerShell executed.

3. Types of Threats and Their Sources

One possible threat is unauthorized remote command execution. Its source may be an external attacker who has obtained the credentials of an account permitted to connect through WinRM. If the connection succeeds, the attacker may run commands on another computer under that account.

A second threat is the misuse of administrative privileges. Its source may be a compromised administrator account or a person who deliberately uses their access outside their assigned duties. Administrative permissions can allow actions that affect other computers and users.

A third threat is movement between computers within a network. An attacker who has already gained access to one account or system may try to use WinRM to reach another system. In this case, remote access may help the attacker expand their presence in the network.

These scenarios have different sources, but they can look similar to ordinary administration.

PowerShell and WinRM are legitimate tools; their use alone does not identify an attacker. The circumstances must be examined before classifying an event as malicious.

4. MITRE ATT&CK Mapping

Our topic is related to two MITRE ATT&CK techniques. T1059.001, "Command and Scripting Interpreter: PowerShell," describes the use of PowerShell for executing commands and scripts.

T1021.006, “Remote Services: Windows Remote Management,” describes the use of WinRM to access a remote system.

The two techniques cover different parts of the activity we study. WinRM provides access to the remote computer, while PowerShell is used to execute commands. This mapping helps us describe a possible attack, but it does not mean that every use of these tools is malicious.

5. Conclusion

In Week 1, we identified the technical terms needed for our topic and classified three possible threats: unauthorized remote command execution, misuse of administrative access and movement between computers through WinRM. The main issue for further investigation is how to distinguish these activities from authorized administrative work.

In Week 2, we will examine public sources for information about the use of remote PowerShell in attacks and record where each finding came from.

References

Microsoft Learn. Security considerations for PowerShell Remoting using WinRM.

<https://learn.microsoft.com/en-us/powershell/scripting/security/remoting/winrm-security>

Microsoft Learn. Running Remote Commands.

<https://learn.microsoft.com/en-us/powershell/scripting/security/remoting/running-remote-commands>

Microsoft Learn. Invoke-Command.

<https://learn.microsoft.com/powershell/module/microsoft.powershell.core/Invoke-Command>

MITRE ATT&CK. Command and Scripting Interpreter: PowerShell (T1059.001).

<https://attack.mitre.org/techniques/T1059/001/>

MITRE ATT&CK. Remote Services: Windows Remote Management (T1021.006).

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